

Key: The District That Was Not Drawn

84-355 Democracy's Data | Vote dilution, Houston County, Georgia | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure. The brief (`vote-dilution-brief.Rmd`) and the lab (`vote-dilution.Rmd`) run the same sequence — the brief in prose with D3 figures, the lab with the code exposed. **Students should read the brief first.** The brief numbers its Steps 1–10; the lab numbers the same material Parts 1–10. They correspond one to one.

This is the lab where a confident conclusion is produced, held for two sections, and then demolished by a fact that was on the table the whole time. The demolition is the point. Do not let Part 7 arrive as a footnote.

The law under it moved on 29 April 2026, and unlike the companion RPV lab, here the change lands on the *evidence-production* step rather than the estimation step. *Louisiana v. Callais*, 608 U.S. ____ (2026), left *Gingles* standing and “update[d]” it. Precondition one now carries two conditions: a plaintiff “cannot use race as a districting criterion” in an illustrative map, and the map “must meet all the State’s legitimate districting objectives, including traditional districting criteria and the State’s specified political goals,” achieving them “just as well” as the State’s own map (slip op., at 29). **The lab’s dial is a machine for violating the first condition, deliberately, and it cannot even be evaluated against the second.** Both facts are marked in the lab and the brief. See *Callais and what it does to this lab*, below.

Every number below was computed from the built files in `data/`.

The session in one sentence

A claim about a district that does not exist can only be settled by building one — so the evidence is manufactured by the party that needs it, every choice the builder makes moves the answer, and a search that fails proves nothing at all.

The findings

```

c(blocks      = nrow(b),
  population   = POP,
  black_adults_pct = r2(SH_VAP),
  black_reg_pct  = r2(SH_REG),
  deviation_spread = r2(DEV_RANGE),
  large_enough_by = r2(TIMES),
  dissimilarity  = DISS,
  reach1_adults_pct = r2(R1$vap_black_pct),
  reachInf_adults  = r2(RI$vap_black_pct),
  enacted_d4_adults = r2(E4$vap_black_pct),
  enacted_d4_pp    = round(E4$pp, 3),
  algo_beating_d4  = BEAT)

```

```

#>      blocks      population black_adults_pct      black_reg_pct
#>      3269.000      163633.000           32.430           35.570
#> deviation_spread large_enough_by      dissimilarity reach1_adults_pct
#>           1.650           3.240           39.880           41.270
#> reachInf_adults enacted_d4_adults      enacted_d4_pp      algo_beating_d4
#>           60.000           53.730           0.266           0.000

```

- **The enacted plan spans 1.65 points against a 10-point allowance.** Malapportionment is over in one line. That is the setup.
- **“Sufficiently large” is satisfied 3.24 times over** — 39,605 Black adults where a bare majority of one district needs 12,212.
- **The dial runs from 41.27% Black adults at Polsby-Popper 0.324 to 60.00% at 0.008.** Buying 18.7 points of Black share costs a factor of 41 in compactness.
- **Enacted District 4 is 53.73% Black among adults at Polsby-Popper 0.266.** 4 of 15 algorithm districts are more Black; 2 are more compact; 0 are both.
- **Where you start barely matters — until race stops driving.** At unlimited reach, 306 starting blocks span 2.08 points. At `reach = 1` the same 306 starting blocks span **22.64 points**. This is new to the key and it is the best thing in it — see Option B.

The three things to get said

1. Failing to find something is not evidence that it is not there. Part 6 produces an honest, careful, quantified conclusion — *a majority-Black district here requires abandoning compactness* — and Part 7 exhibits a district that is both more Black and more compact than anything the search produced, and that has been enacted law for years. **Make somebody articulate the asymmetry:** exhibiting a map settles the question in the affirmative permanently; failing to find one settles nothing, because no search of this kind is ever exhaustive. Precondition one is provable and not disprovable, and that structural fact is what the fight over “what counts as a demonstration map” is really about.

2. The choice that moves the answer is the one that looks like a setting. Part 9 is deliberately anticlimactic: at the dial’s far end the seed does almost nothing (2.08 points across 306 restarts) and the dial does everything. `reach` is not a statistical parameter with a correct value — it encodes *what you think a district is allowed to look like*, which is a legal judgment wearing the clothes of a tuning knob. **The lesson is not “analyst choices matter.” It is that they matter unequally, and the important one was disguised.**

Say the scope out loud, because Option B breaks it: *at unlimited reach* the seed is nearly irrelevant, and at `reach = 1` it is worth 22.64 points. **Which knob matters is itself contingent on where the other**

knob is set, so “we checked robustness to the starting block” is a claim about one configuration and not about the method.

3. “Majority” has three denominators and they disagree. Part 8 puts up 12 four-district configurations in which **every single one is majority-Black among registered voters and not one is majority-Black among voting-age population.** Same blocks, same lines, same people. *Bartlett v. Strickland* says a real numerical majority is required and does not say of what. **Ask the room to pick before revealing that both answers are defensible.**

The line for the board: the map that does not exist is the one the case turns on, and somebody has to draw it before anyone can argue about it.

Walking the steps

Part	What it does	Min	Watch for
1 What you observe	Three Black shares, three denominators	5	Point at the three numbers now. Part 8 cashes them.
2 Dilution by subtraction	1.65 points against 10	5	Students expect this to be the lab. It is the setup.
3 What that rule replaced	County unit rule on 2020 population	5	228-to-1. The rule worked so well it stopped being the argument.
4 The half you cannot subtract	Large enough 3.24× over; dissimilarity 39.88	7	Large and spread evenly beats small and clustered. Not intuitive.
5 One algorithm, one dial	grow() , and the seed	8	Read the seed line aloud. Race picked the starting block.
6 Turning the dial	The frontier, and the confident conclusion	11	Take the prediction before the table. Let them commit.
7 The map already exists	Enacted D4 dominates everything	10	The pivot. Do not preview it. Do not rush it.
8 “Majority” of what	12 districts, two answers	6	Take a vote on the denominator first.
9 Does the seed matter	2.08 points across 306 restarts	4	Anticlimax on purpose. Then Option B breaks it.
10 Your turn	Five options	4	Steer strong students to B or C.

Part 6 deserves the prediction, and it is the one prediction in this course that almost everybody gets wrong in the same direction. Before the table, ask the room for two numbers: the highest Black share a *contiguous* district of 32,727 people can reach, and how compact that district will be. The county is

32.43% Black among adults and holds 3.24 times the Black adults a majority district needs, so most students guess something comfortably over 50% at a respectable shape. The honest answer at a respectable shape is 41.27%, and the first setting that clears 50% — **reach** = 40 — does it at Polsby-Popper 0.036, about 9 times worse than where it started.

If you are short on time, protect Parts 6, 7 and 8. The confident conclusion, its demolition, and the denominator. Parts 2 and 3 can be compressed to four minutes together with no loss.

The pivot, and how to run it

Do not tell them the enacted map is coming. The lab is written so Part 6 reads as a finding and Part 7 reads as a correction, and the sequence only works if the room has actually accepted Part 6. Three moves:

1. **Make them write the sentence.** At the end of Part 6, have someone say out loud what an expert would put in a report: *in Houston County a majority-Black district can be constructed only by abandoning compactness*. Nothing in the analysis up to that point is wrong. The arithmetic is right, the frontier is real, the compactness measures are standard.
2. **Then run Part 7 without commentary.** Print the table. Let them find District 4 in it. 53.73% at Polsby-Popper 0.266 — better than every setting on both axes at once.
3. **Then ask what went wrong.** The productive answers are all about the search and none about the county: it grows from a single seed, it never backtracks, it is greedy, and it only ever builds one district. The unproductive answer is “the algorithm was bad.” It was a perfectly ordinary algorithm, and the frontier it traced is a **lower bound** on what a mapmaker can do — which is all any such search can ever produce.

The wobble is corroborating evidence and worth two minutes. At **reach** = 2 the district *drops* to 41.14%, below several tighter settings. A greedy search is not guaranteed to be monotone in its own parameter. **A student running Option A at k = 150 will find 49.23%** — also below **reach** = 100’s 55.18%. So it is not one bad point; it is a whole region where reaching *further* finds *less*. The curve is a property of the search, not of the county, and that is exactly the point Part 7 makes geometrically.

Say what Part 7 does not show. It is not a claim about how the enacted plan was drawn or with what in front of whom — that question is not in this file. And it is not a comparison of two illustrative maps: an enacted plan and a plaintiff’s demonstration exhibit are different objects judged by different rules. The claim is about evidence, and only about evidence.

Callais and what it does to this lab

The lab got better, not worse, and it is worth saying why. Before 29 April 2026 the dial was a nice illustration of researcher degrees of freedom. Now every setting except the leftmost describes a practice the Court has disqualified, so the dial is also a dated artefact — and students can watch a methodological choice become a legal one without any of the underlying numbers changing.

Three things to have straight before the room asks.

- **The first condition bites on the dial.** Plaintiffs “cannot use race as a districting criterion.” Every point on the Part 6 frontier except **reach** = 1 was produced by letting race choose the next block.

- **The seed is not exempt, and this is the good question.** `reach = 1` takes race out of the *annexation rule* and leaves it in the *starting block*, which is `which.max(b$vap_black)`. If race may not be a districting criterion, a race-chosen seed is a problem too. **Option B is where a student can measure what that choice is worth**, and the answer is: at `reach = 1`, nearly everything.
- **The second condition is the one nothing here can meet.** An illustrative map must satisfy all the State’s legitimate objectives — traditional criteria and “the State’s specified political goals” — “just as well” as the enacted map. That needs a complete partition of the county, a statement of Georgia’s stated objectives, and election results. **This folder has none of the three.** If a student concludes that `reach = 1` is therefore “the admissible answer,” that is a 3, not a 4: `reach = 1` clears the first condition and is untested against the second.

And the sequel is worth thirty seconds. Five weeks after *Callais*, the Court stayed the renewed injunction in the *Milligan* litigation itself, on the ground that the plaintiffs’ map had not performed “just as well” on Alabama’s stated criteria of keeping the Gulf Coast together and avoiding incumbent pairings. *Allen v. Milligan*, 608 U.S. ____ (2026) (per curiam) (June 2, 2026). **The plaintiffs who won in 2023 no longer have their district.** Students who read *Milligan* as the stable point of this area of law should hear that.

Handling the room

- **Nobody writes code here.** Say it at the top, as always. The `grow()` function is fifteen lines and is printed so that students can see there is no magic in it, not so they can write one.
- **Somebody will say the exercise is racist.** Take it seriously rather than deflecting, because the observation is correct and the lab is built on it: to find out whether Black voters *could* have a district, the algorithm sorted every block in the county by race. Section 2 remedies require sorting voters by race in order to stop voters being sorted by race. That tension is the subject, it is why *Shaw* and *Miller* exist, and it is what *Callais* claims to resolve. **There is no race-blind way to ask this question**, and saying so plainly is better than any answer that pretends otherwise.
- **The live case needs one sentence, early.** The instructor is retained by plaintiffs in *Driver v. Houston County*, No. 5:25-cv-00025 (M.D. Ga.). No expert-report material is used, every non-enacted district here was drawn by the printed algorithm, and the lab reaches no legal conclusion. Say it once, at the start, and move on.
- **Compactness scores are not facts about virtue.** A district bounded by a winding river scores badly for reasons no mapmaker chose, and any district can be made to score better by smoothing a boundary without moving a single voter.

Option A — Move the dial yourself

Verified output

```
t(sapply(c(5, 30, 150), function(k) {
  g <- grow(seed, IDEAL, k)
  c(reach = k, blocks = length(g), r2(score(g)))
}))
```

```
#>      reach blocks population adults registered
#> [1,]      5      579      47.93 43.69      48.89
#> [2,]     30     614     52.26 48.25     52.84
#> [3,]    150     781     52.15 49.23     52.09
```

For comparison, the settings on either side, from `tradeoff.csv`:

```
print(t5[t5$reach %in% c(4, 6, 24, 40, 100, 200),
      c("reach", "vap_black_pct", "reg_black_pct", "pp", "reock")],
      row.names = FALSE)
```

```
#> reach vap_black_pct reg_black_pct    pp reock
#>    4      43.575      48.851 0.202 0.769
#>    6      42.828      48.917 0.174 0.727
#>   24      47.692      52.275 0.074 0.599
#>   40      51.070      55.569 0.036 0.336
#>  100      55.179      58.859 0.023 0.309
#>  200      42.680      41.837 0.006 0.248
```

3/4: Runs a value of *k*, reports the Black share, and places it correctly between its neighbours on the Part 6 curve.

4/4: Answers the second half of the question — could you defend that *k* to a judge who asked why you chose it? — and concludes that you could not, because `reach` has no meaning outside this program. There is no *k* that corresponds to a districting principle, which means the whole curve is a family of arbitrary maps and the analyst picks a point on it.

4/4, best: Picks *k* = 150 (or any value in the 100–200 range) and notices that it comes out at **49.23%**, below `reach` = 100’s **55.18%** — so the wobble the lab flags at `reach` = 2 is not an isolated point but a region. A student who works out *why* a greedy search can do worse with more options (early annexations foreclose later ones; the centroid drifts and changes what counts as “near”) has understood that the frontier is an artefact of the procedure. That is the same lesson as Part 7, reached independently.

Option B — Start somewhere else

This is the most valuable option in the lab and the one to steer good students toward. It produces a result that contradicts the lab’s own Part 9, and the contradiction is correct.

Verified output

```
t(sapply(c(1, 10, 50), function(i) {
  s <- ord[i]
  c(rank = i, seed_black_adults = b$vap_black[s],
    reach1_adults = unname(r2(score(grow(s, IDEAL, 1))[2])),
    reachInf_adults = unname(r2(score(grow(s, IDEAL, Inf))[2])))
}))
```

```
#>      rank seed_black_adults reach1_adults reachInf_adults
#> [1,]      1             413          41.27          60.00
#> [2,]     10             245          43.02          60.29
#> [3,]     50             109          42.69          60.00
```

The full sweep — the same 306 starting blocks the build script used, re-run at `reach = 1`:

```
c(seeds          = length(R1SW),
  reach1_min     = r2(min(R1SW)),
  reach1_max     = r2(max(R1SW)),
  reach1_sd      = r2(sd(R1SW)),
  reach1_spread  = r2(R1SW_SPREAD),
  reachInf_spread = r2(SW_SPREAD),
  reachInf_sd    = r2(sd(sw$d5_vap_black_pct)),
  reach1_over_50 = sum(R1SW > 50))
```

```
#>      seeds      reach1_min      reach1_max      reach1_sd      reach1_spread
#>    306.00         27.21         49.85         8.77         22.64
#> reachInf_spread reachInf_sd reach1_over_50
#>         2.08         0.48         0.00
```

Part 9 says the seed barely matters. Part 9 is measuring the race-aware algorithm. At unlimited reach the seed is nearly irrelevant — 2.08 points across 306 restarts — because the search reaches far enough to find the same Black population wherever it starts. At `reach = 1` the same 306 restarts span **22.64 points**, from 27.21% to 49.85%, a standard deviation of 8.77 against 0.48.

3/4: Runs the three seeds, reports that the answers differ more at `reach = 1` than the lab’s Part 9 suggested.

4/4: Explains the mechanism. Unlimited reach can go and get the Black population from anywhere in the county, so the starting block washes out; at `reach = 1` the district is whatever happens to be adjacent, so it is a description of one neighbourhood rather than a search. **The constraint that makes the map legally admissible is the same constraint that makes it depend almost entirely on an arbitrary choice.**

4/4, best: Closes the loop on *Callais*. The seeds were ranked by `order(-b$vap_black)` — chosen by race. So the one race-based decision left in the `reach = 1` construction is now the decision that determines the answer, and it moves it 22.64 points. A student who asks whether a `reach = 1` map grown from a race-chosen seed is really a race-blind map, and notices that **none of the 306 race-blind-growth districts clears 50% anyway** (the best is 49.85%), has produced a better argument than the lab makes on its own. Give it a 4 and use it next year.

Option C — Change the denominator

Verified output

```
rbind(
  `reach 1, max adults`      = r2(score(CA1)),
  `reach 1, max registered`  = r2(score(CR1)),
  `reach Inf, max adults`    = r2(score(CAI)),
  `reach Inf, max registered` = r2(score(CRI)))
```

```
#>               population adults registered
#> reach 1, max adults      45.34  41.27    48.17
#> reach 1, max registered  45.34  41.27    48.17
#> reach Inf, max adults    62.09  60.00    58.04
#> reach Inf, max registered 56.57  53.87    66.59
```

```
identical(sort(CA1), sort(CR1))      # reach 1: the very same blocks
```

```
#> [1] TRUE
```

```
r2(JAC)                             # reach Inf: % of blocks shared
```

```
#> [1] 78.59
```

The reach = 1 rows are identical, and that is the finding. With one candidate on the frontier there is no choice to make, so the objective function never runs. **The denominator only matters when the dial lets it matter** — which means the two analyst choices in this lab are not independent, and a student who sees that has spotted something the brief does not say.

At unlimited reach they separate hard. Maximising adults gives 60.00% of adults and 58.04% of registrants; maximising registrants gives 66.59% of registrants and only 53.87% of adults. The two districts share 78.6% of their blocks.

3/4: Swaps the columns, reports that the district changes, and notes the Black share is higher on the registered measure.

4/4: Answers the normative half. Courts generally use voting-age or citizen-voting-age population, and the case for it is that registration is *endogenous to the wrong being alleged* — if a group is discouraged from registering, using registration as the denominator lets the discouragement manufacture the majority. The case for registration is that it measures the actual electorate, and *Bartlett's* concern was whether a group can elect its candidate, which registrants do and non-registered adults do not.

4/4, best: Notices the reach = 1 identity above and draws the general lesson: **a sensitivity analysis that varies one choice at a time will report that the denominator does not matter, because at the defensible setting of the other knob it genuinely does not.** That is a real methodological trap and it is worth naming for the whole room if a student finds it.

Option D — Four districts instead of five

Verified output

```
rbind(`reach 1` = r2(D41),
      `reach Inf` = r2(D4I))
```

```
#>               population adults registered
#> reach 1      48.21  44.33    50.37
#> reach Inf    58.52  56.53    54.97
```



```
print(f4[, c("reach", "pop_black_pct", "vap_black_pct", "reg_black_pct")],
      row.names = FALSE, digits = 4)
```

```
#> reach pop_black_pct vap_black_pct reg_black_pct
#>    1         48.21         44.33         50.37
#>    2         47.92         44.05         50.08
#>    3         49.32         45.31         50.57
#>    4         49.57         45.57         50.71
#>    6         49.25         45.38         50.77
#>    8         49.88         45.91         51.03
#>   12         50.67         46.78         51.71
#>   16         50.77         46.84         51.88
#>   24         52.19         48.30         53.20
#>   40         52.75         48.93         53.98
#>   60         53.06         49.28         54.07
#>  100         53.12         49.59         53.73
```

Larger districts, and the arithmetic gets easier: the ideal rises from 32,727 to 40,908 people. But **the answer depends entirely on the denominator, which is the trap the option is built around.** Across all 12 finite settings at or below `reach = 100`, the Black share of registered voters never falls below 50.08% and the Black share of adults never rises above 49.59%. Every one of them is majority-Black on one measure and not the other.

3/4: Re-runs with `ideal4`, reports that the shares are higher than in the five-district case, and concludes a majority-Black district is easier.

4/4: Refuses to answer “easier or harder” without naming the denominator, because on registered voters the four-district configuration clears 50% at every finite setting including `reach = 1`, and on voting-age population it clears 50% at none of them.

4/4, best: Notices that fewer, larger districts is not a neutral change — it is a change in the *unit of representation*, and it makes proportionality coarser in a way that cuts both directions. A group at 32.43% of adults is entitled by proportionality to 1.62 of five seats and 1.30 of four. **Shrinking the board makes each seat worth more and makes the group’s claim to a whole one weaker**, even as it makes the district easier to draw. Reward anyone who gets both halves.

Option E — Your own question

Grade on whether the question could be answered with what is in the folder.

Good ones that have come up:

- *Does the seed’s own Black share predict the district’s?* It does at `reach = 1` and barely at all at unlimited reach — a clean restatement of Option B.
- *How many blocks does each district use?* `reach = 1` takes 540 blocks to reach the ideal population; unlimited reach takes 1528 — 2.8 times as many for the same people. That ratio is what a tendrill looks like numerically.
- *What if contiguity is dropped entirely?* That is the ceiling already in the lab: 67.10% at Polsby-Popper 0.006. A student who asks why the ceiling is not simply the answer has understood that “could a district be drawn” means a district somebody could enact.

Questions the folder cannot answer, which should be redirected rather than penalised: anything about how anyone voted (no election results are here), anything about citizenship (registration is not citizenship), and anything requiring a complete five-district plan.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is given, and it runs.
- **The move that earns a 4** is treating the failed search in Part 6 as information about the search rather than about the county. Everything else in the lab is downstream of that distinction.
- **A 2** is “the algorithm was bad” or “you can make data say anything.” Both skip the work. The algorithm was ordinary and the data were not manipulated; that is why the result is disturbing.
- **Also a 2:** reporting a number off the Part 6 frontier as though it were a fact about Houston County. The frontier is a lower bound.
- **A 3, not a 4:** concluding that `reach = 1` is “the answer after *Callais*.” It satisfies the first condition and is untested against the second, and its own value swings 22.64 points on the seed.
- **Reward anyone who asks what a *complete plan* would have shown.** The algorithm builds one district and stops; a real demonstration plan assigns every block and holds all five inside the population band at once. Noticing that the lab answers an easier question than a court would ask is the most sophisticated available response, and it is not stated anywhere in the lab except in the limitations.

Anticipated questions

- “*Why not just look at the enacted map from the start?*” — Because in a real case the map you need does not exist; that is the entire difficulty. The enacted plan is available here as an answer key, and answer keys are what this course uses to grade methods that are normally unfalsifiable. Same move as the BISG and RPV labs.
- “*Isn’t Polsby-Popper just a bad measure?*” — It has real defects, and Duchin and Tenner (2024) argue the whole contour-based family shares them. But the qualitative result here does not depend on the measure: Reock moves the same way, from 0.766 to 0.284. Two contour measures are one limitation, not two independent checks — say that too.
- “*Could a second majority-Black district be drawn?*” — Unknown, and the lab cannot say. The algorithm builds one district and stops. A complete plan can fail for reasons having nothing to do with the group, such as running out of population in the right places.
- “*Is `reach = 1` what an expert would actually file now?*” — No. It is the only end of *this* dial that survives the first *Callais* condition, which is not the same as being an admissible demonstration map. Nobody yet knows what the procedure is; the Court required the showing and did not specify how to make it.
- “*The registration file has race on it — why is the census used at all?*” — Because registration is endogenous to the claim and the census is not, and because *Bartlett* is generally read against voting-age population. Part 8 is the demonstration that this is a choice and not a technicality.
- “*Did we prove Houston County dilutes anyone’s vote?*” — No, and do not let this become a finding about the county. Nothing here touches preconditions two and three or the totality of the circumstances. **Precondition one is a threshold, not a verdict.**

Connections

- **rpv** — preconditions two and three, same county, same registration file. Run it first; this lab assumes the room already knows what a *Gingles* precondition is. The two labs also split *Callais* cleanly between

them: the RPV lab gets the party-control requirement, this one gets the illustrative-map conditions.

- **big-check and surnames** — where the self-reported race on the registration file comes from, and what the any-part-Black definition costs.
- **areal-units** — districting is the zoning effect deployed on purpose. 3,269 blocks re-bundled into five districts give different summary numbers depending only on where the lines fall; that lab is the general case and this one is the consequential instance.
- **precinct-geography** — the other lab where two defensible methods on the same data disagree about something basic. Pair the two if you want the methodological point to land twice.
- **apportionment** — the other place a technical rule with no obvious politics turns out to allocate something people care about, and the other place equal population is the whole argument.